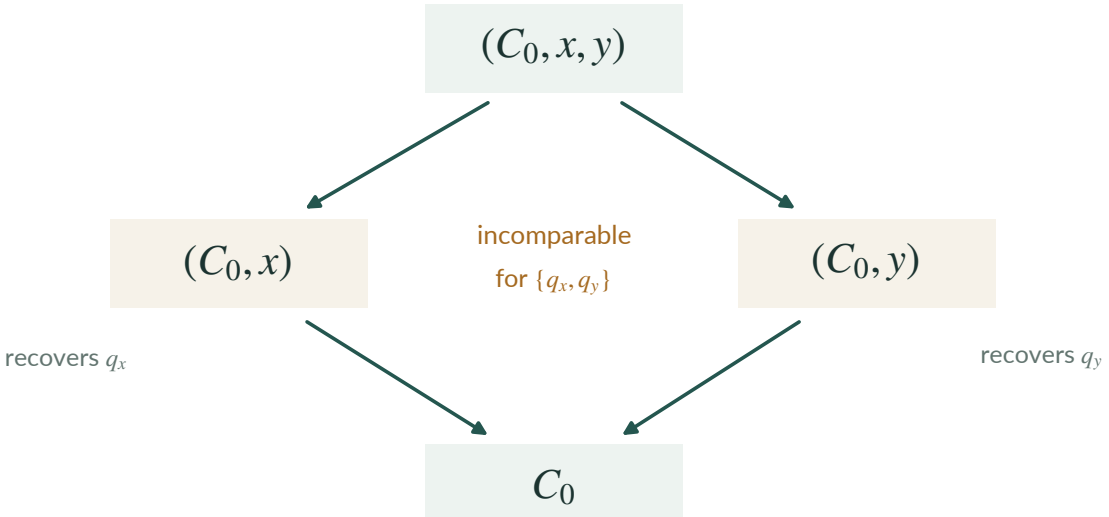

Arrows discard information. The two middle schemes are incomparable for $\{q_x, q_y\}$.